

Phase 2 Evaluation — The AI Bidding War: Autonomous Procurement and Vendor Negotiation

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Track: A (technical build)

Score: 98 / 100

Dimension	Score	Max
Architecture and role design	25	25
Tools, memory, data, and state design	19	20
Prototype quality	15	15
Evaluation plan and test design	19	20
Risk and governance thinking	10	10
Documentation and contribution update	10	10
Total	98	100

Dimension Notes

Architecture and role design — 25/25. The hub-and-spoke design is tied directly to the problem it solves, and the three-layer separation of negotiation, assurance, and execution is justified rather well. Role definitions are complete for all seven components, coordination logic names four explicit termination conditions, and the rule that a governance nudge applies only while vendor rounds remain is a precise piece of design work. The comparison to a single-thread baseline explains why the simpler alternative fails for this specific task. Inspection of `orchestrator.py` and `vendor_histories` confirms the isolation is structural rather than prompt-level.

Tools, memory, data, and state design — 19/20. The per-role information access table states what each role can and cannot read. The tools and memory matrix separate configuration, LLM access, deterministic execution, in-session state, audit serialization, and persistence into distinct layers with file paths. The three state objects `vendor_histories`, `conversation_buyer`, and `buyer_reply_prev` are each specified in terms of what they contain and when they update. The one small gap is that retention and rotation of the logs directory is not discussed, which matters if this ever moves out of lab scope.

Prototype quality — 15/15. The prototype runs end to end across eleven scenarios. Scenario logs of 800 to 900 lines each show the full negotiation, including isolated vendor prompts, Strategic Audit output, human approval or veto, and guardrail result. Scenario 2 alone records three separate human vetoes of over-ceiling proposals, which is the kind of evidence that makes a governance claim testable. Both `SUCCESS` and `NO_AWARD` paths are instrumented, and the evidence logs are structured JSON that a grader can query directly.

Evaluation plan and test design — 19/20. The two-layer scoring of governance versus negotiation prevents conflating policy enforcement with procurement outcome, and every

scenario has a measurable pass condition with numeric thresholds rather than a descriptive one. Adversarial cases are planned into the design, not patched in. Scenario 2 is a deliberate fail-by-design, Scenario 6 tests compromised narrative framing against a hard cap, Scenario 10 tests bid steering, and Scenario 11 tests conflict of interest. Three distinct failure-mode writeups separate structurally different governance mechanisms that all happen to produce NO_AWARD. The one remaining gap is the absence of a multi-seed variance study, which is acknowledged as a deliberate scope boundary but will matter in Phase 3.

Risk and governance thinking — 10/10. The risk register names six specific failure modes and maps each mitigation to a named architectural component rather than a generic principle. The deterministic guardrail outside the LLM is implemented as advertised, with BUYER_HARD_LIMIT set to 15000 and no model call in the award path. The decision not to auto-inject the full Strategic Audit into the buyer's next turn shows attention to the negotiation-versus-assurance boundary, and the rule that human rejection on finalize does not trigger a nudge shows the same discipline applied to the round-versus-finalize boundary. Trust-boundary thinking is present throughout, not bolted on.

Documentation and contribution update — 10/10. The packet is organized so a reviewer can locate any required element quickly: separate Phase 1, Phase 2, and Phase 3 folders, agent and multi-agent canvases, evaluation CSVs with a failure log, a supplementary evaluation report, a standalone AI usage disclosure, a README, a requirements file, and optional Docker packaging. Terminology is consistent across narrative, code, canvases, and logs. The AI usage disclosure goes beyond the minimum by separating human-led decisions from coding-agent assistance. Solo project, so contribution attribution is inherent.

What worked

The three Phase 1 action items are all addressed with evidence. The ambiguous-scenario pass criteria are now numeric, the Strategic Advisor has its own first-class canvas entry with inputs, outputs, and failure mode, and state management across rounds is specified both in the narrative and in the code. The Phase 1 section in the supplementary evaluation report explicitly names what changed in response to each piece of feedback, which closes the feedback loop.

The second clear strength is the separation of governance and negotiation in the evaluation design. Treating these as two independent scoring layers means the submission can honestly report that all eleven scenarios pass governance while one fails negotiation by design and another is partial. That discipline protects the integrity of the claims being made.

Three things to address in Phase 3

1. Run multi-seed variance evaluation on the scenarios that currently rely on a single harvest. The Phase 3 evidence package expects at least five completed scenarios with results and at least two failure cases, and you already exceed that count. What the current harvest does not show is whether the outcomes are stable under LLM stochasticity. Pick the scenarios where the outcome was borderline, specifically Scenario 4 and Scenario 5 where the final award landed exactly at the 15000 ceiling and run each at least three to five times. Report outcome distributions rather than point values. Without this, Phase 3 claims about reliability will be harder to defend.

2. Add a failure case that is not by design, meaning a case where the system behaved worse than expected under ordinary conditions. The three current failure writeups are all cases where NO_AWARD is the correct governance outcome. Phase 3 asks for at least two failure cases, and the rubric rewards honest acknowledgment of unresolved gaps. Find or construct a scenario where the audit missed something, or where the buyer's negotiation strategy produced a weak deal that was technically in-budget, or where the orchestrator handled an edge case such as an API timeout or malformed BID_JSON in a way you would not ship. Document it as FL-004 with the same template. This converts the failure-case section from a governance demonstration into evidence of where the system has real limits.

3. Write the 5-minute video script around the three-layer separation as the core claim, with one scenario as the demonstration. The submission has eleven scenarios and three failure cases, which is more than fits into five minutes. Pick one scenario that shows the governance chain end to end, ideally Scenario 2 with its three vetoes, and use it to illustrate how negotiation, assurance, and execution are structurally separate. Everything else in the evidence package should be pointed to, not shown. Without this discipline, the video will either rush through material or leave the central argument implicit.